

Computer Programming

Lab 10 Lab Title

More About Functions in C++

Objective

The objective of this lab is to learn and apply:

1. Function Call by Value
2. Function Call by Reference
3. Default Values in Function Parameters

Software and Tools Required

Software	Purpose
Dev-C++	Writing and running C++ programs
Code::Blocks	Alternative C++ IDE GNU Compiler
Collection Program compilation	

Installation Procedure

Installing Dev-C++

Step 1

Download and install Dev-C++ from the official website.

Step 2

Open the installer and click Next.

Step 3

Accept the license agreement.

Step 4

Choose installation location and click Install.

Step 5

After installation, open Dev-C++.

Theory

1. Call by Value

Software and Tools Required

Theory

1. Call

- In call by value, a copy of the variable is passed to the function.
- Changes made inside the function do not affect the original variable.

Formula Representation

$x_{copy} = x_{\{copy\}} = x$

Advantages

- Safe method
- Original data remains unchanged

Disadvantages

- Extra memory is used

2. Call by Reference

- In call by reference, the address of the original variable is passed to the function. Changes inside the function affect the original variable.

Reference Representation

`f(&x)f(\&x)f(&x)` Advantages

- Faster execution
- Original values can be modified

Disadvantages

- Original data may change accidentally

3. Default Parameters

- Default parameters are values assigned to function parameters during declaration.
- If no argument is passed, the default value is used automatically.

Example:

`divide(a=8,b=2)=4``divide(a=8,b=2)=4``divide(a=8,b=2)=4`

Lab Task 1

Function with Parameters

Problem Statement

Create a function named `multiply` that:

- Takes two integer parameters
- Multiplies them □ Displays the result

Algorithm

1. Start
2. Declare function `multiply()`
3. Input two integers
4. Multiply numbers
5. Display product
6. End

Program Code

```
#include <iostream> using
namespace std;

// Function declaration void
multiply(int num1, int num2);
int
main()
{
    // Function call
    multiply(5, 4);
    return
0;
}

// Function definition void
multiply(int num1, int num2)
{    int
product;

    // Calculating multiplication
product = num1 * num2;

    // Displaying result
cout << "Product = " << product << endl; }
```

Output

Product = 20

Program Explanation

- multiply() function receives two integers.
- Product is calculated using * operator.
- Result is displayed inside the function.

Lab Task 2

Function with Return Value

Problem Statement

Create a function named `getSquare` that:

- Takes one integer parameter
- Finds square of number
- Returns the result

Algorithm

1. Start
2. Declare function `getSquare()`
3. Input number
4. Calculate square
5. Return result
6. Display answer
7. End

Program Code

```
#include <iostream> using
namespace std;

// Function declaration int
getSquare(int number);
int main() {
    int result;
    // Function call
    result = getSquare(6);

    // Displaying result      cout <<
    "Square = " << result << endl;
    return
    0;
}

// Function definition int
getSquare(int number)
{
    // Returning square
    return number * number;
}
```

Output

Square = 36

Mathematical Representation

$\text{Square} = n^2$

Explanation

- `getSquare()` accepts one integer.
- Square is calculated by multiplying number with itself.
- Function returns the value to `main()`.

Conclusion

In this lab:

- We learned how functions work in C++.
- We studied call by value and call by reference.
- We used default parameters.
- We created functions with parameters and return values.

Functions help make programs:

- Simple
- Reusable
- Organize

Precautions

- Write correct syntax.
- Use semicolon ; carefully.
- Declare functions before calling.
- Use proper indentation.
- Compile program before execution.